FACULTY OF BUSINESS SCIENCES

DEPARTMENT OF INFORMATION AND MARKETING SCIENCES

NAME: CHISORO SIMBARASHE

REGISTRATION: R259880M

MODULE CODE: DSI143- FUNDAMENTALS OF DATA WAREHOUSE AND DATA MINING

PROGRAM: DATA SCIENCE AND INFORMATICS (HARARE WEEKEND CLASS)

ASSIGNMENT 1

LECTURER: K. CHINYONGO

Project available on

https://github.com/chisoro/dataWareHouse

Project Plan on

https://www.notion.so/Data-Warehouse-Assignment-3588f4ddd6ad80eab012e8839baa9be3?source=copy_link

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REQUIREMENTS ANALYSIS

Data Warehouse Development (Medallion Architecture)

- **Objective:** Build a SQL-based, non-historised (truncate/load) data warehouse integrating ERP and CRM CSV data using a Medallion (Bronze/Silver/Gold) framework.
- **Data Quality & ETL:** Clean, de-duplicate, and standardize raw data; implement robust ETL pipelines to transform data into a structured star schema.
- **Deliverables:** End-to-end documentation including architecture diagrams, data dictionary/catalog, naming conventions, and validation reports

Business Intelligence and Analytics

- **Objective:** Develop analytical SQL queries against the Gold-layer star schema to drive decision-making.
- **Key Areas:** Generate actionable insights regarding:-
  - Sales performance

- Customer behaviour
- Product trends.

Summary of Scope

Component	**Focus**
Sources	ERP and CRM CSV Files
Methodology	Medallion (Bronze - Silver - Gold)
Strategy	Current state (Truncate/Reload)
Output	Star Schema and Analytics

Layers of the Data Warehouse

	BRONZE LAYER	**SILVER LAYER**	**GOLD LAYER**
Definition	Raw data	Clean and standardised data	Business Ready Data
Objective	For traceability and debugging	Prepared data for analysis	Provides data to be consumed for analytics
Object Type	Tables	Tables	Views
Load Method	Full-load (Truncate and Insert)	Full-load (Truncate and Insert)	None
Data Transformation	None (As-is)	Cleaning Standardisation Normalisation Feature Engineering Enrichment	Integration Aggregation Business logic and Rules
Data Modelling	As-is	As-is	Star Schema

			Aggregation objects
Consumers	Data Engineers	Data Engineers Data Analysts	Data Analysts Business Intelligence

Naming Conversions

Naming conversions are set of rules or guidelines for naming anything in a project from:

- Database
- Schema
- Tables
- Procedures

In this project I will use the camelCase where the first word is lower case and upper case first letter of all subsequent words.

Language: English

Rule: Avoid using reserved words

Table Naming Conversions

Bronze Rules

- All table names must start with source name, and table names must match original tables renamed to match the camelCase.
- <source><entity>
    - <source>: Name of the source system (ERP,CRM)
    - <entity>: Original name of table renamed to match camelCase
    - Example: crmCustomerInfo – Customer information table from crm file.

Silver Rules

- All table names must start with source name, and table names must match original tables renamed to match the camelCase.
- <source><entity>
    - <source>: Name of the source system (ERP,CRM)
    - <entity>: Original name of table renamed to match camelCase
    - Example: crmCustomerInfo – Customer information table from crm file.

Gold Rules

- All table names must be meaningful business aligned names starting with category prefix.
- <category><entity>
    - <category>: The role of the table such as dimension (dim) or fact table (fact)
    - <entity>: Descriptive name of the table aligned with the business domain (customer, product, sales)
    - Example: factSales – Fact table containing sales information.

Column Naming Conversions

Surrogate Keys

- All primary keys in the dimension tables must use the suffix “Key”
- <tableName><Key>
    - <tableName> refers to the name of the table the key belongs to.
    - <Key> indicates that the column is a primary key.
    - Example: customerKey indicating the primary key from dimension table dimCustomers table.

Technical Columns

- All technical columns must be prefix by “dwh” followed by a descriptive name indicating the column’s purpose.

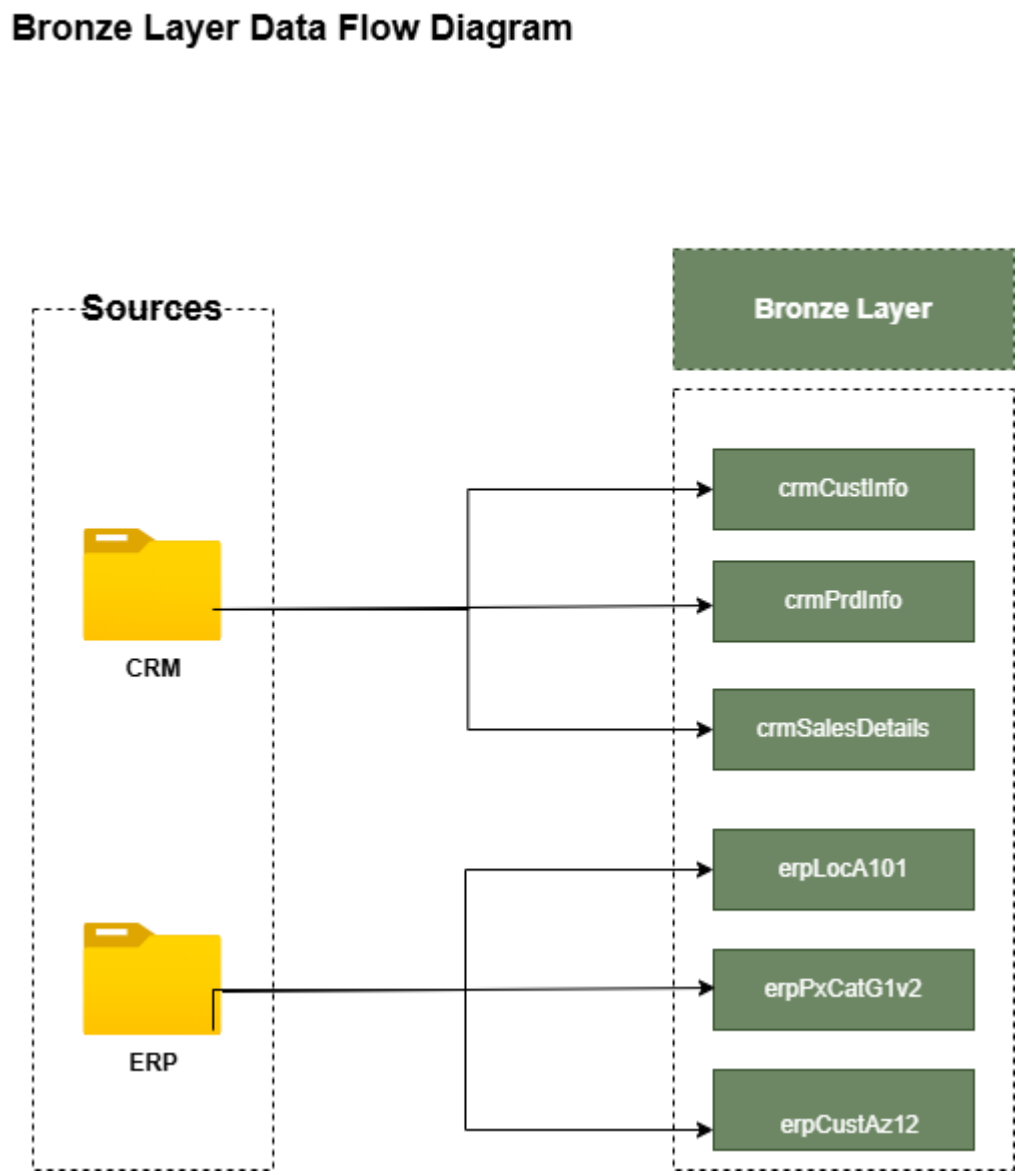
- dwh<ColumnName>
    - dwh: Prefix for system generated data
    - <CloumnName>: Descriptive name indicating column purpose
- Example: dwhLoadDate: System generated column indicating the date when the data was loaded.

Stored Procedures

- All load procedures must have a prefix ‘load’<LayerName>
- Example: loadBronze

Task 1. Bronze Layer

Data Flow Diagrams



Data Analysis and Checks

- Scripts found under the tests folder “checksBronze.sql”

```
==================================================================
Checking bronze.crmCustInfo
==================================================================
Check for NULLs or Duplicates in Primary Key

(6 rows affected)
Check for Unwanted Spaces

(0 rows affected)
Data Standardization & Consistency

(3 rows affected)
```

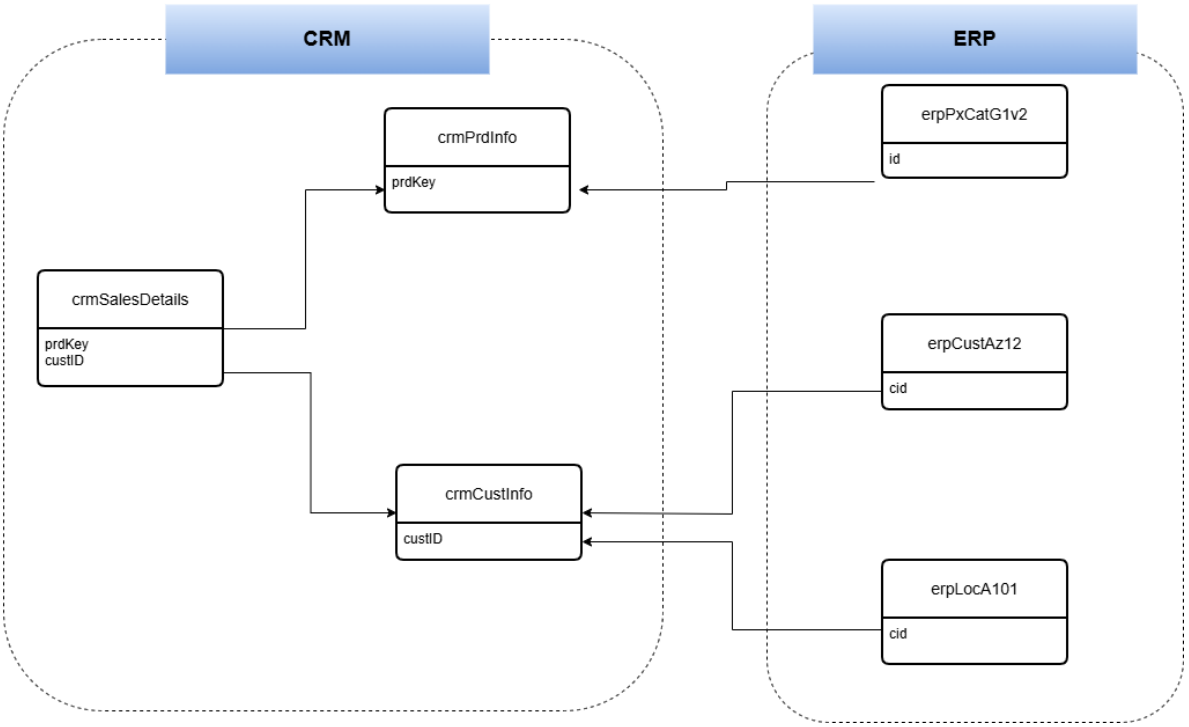
```
=====================================================================
Checking bronze.crmPrdInfo
=====================================================================
Check for NULLs or Duplicates in Primary Key

(0 rows affected)
Check for Unwanted Spaces

(0 rows affected)
Check for NULLs or Negative Values in Cost

(2 rows affected)
Data Standardization & Consistency

(5 rows affected)
Check for Invalid Date Orders (Start Date > End Date)

(200 rows affected)
=====================================================================
Checking bronze.crmSalesDetails
=====================================================================
Check for Invalid Dates

(0 rows affected)
Check for Invalid Date Orders (Order Date > Shipping/Due Dates)

(0 rows affected)
Check Data Consistency: Sales = Quantity * Price

(33 rows affected)
=====================================================================
Checking bronze.erpCustAz12
=====================================================================
Identify Out-of-Range Dates
Expectation: Birthdates between 1924-01-01 and Today. We cannot expect to have a
customer of over 103 years old

(31 rows affected)
Data Standardization & Consistency

(6 rows affected)
=====================================================================
Checking bronze.erpLocA101
=====================================================================
Data Standardization & Consistency

(11 rows affected)
=====================================================================
Checking bronze.erpPxCatG1v2
=====================================================================
Check for Unwanted Spaces

(0 rows affected)
Data Standardization & Consistency

(2 rows affected)

Completion time: 2026-05-08T15:32:08.1387127+02:00
```

The report forms the bases of our transformations, normalisations and data cleaning to transfer data to Silver Layer.

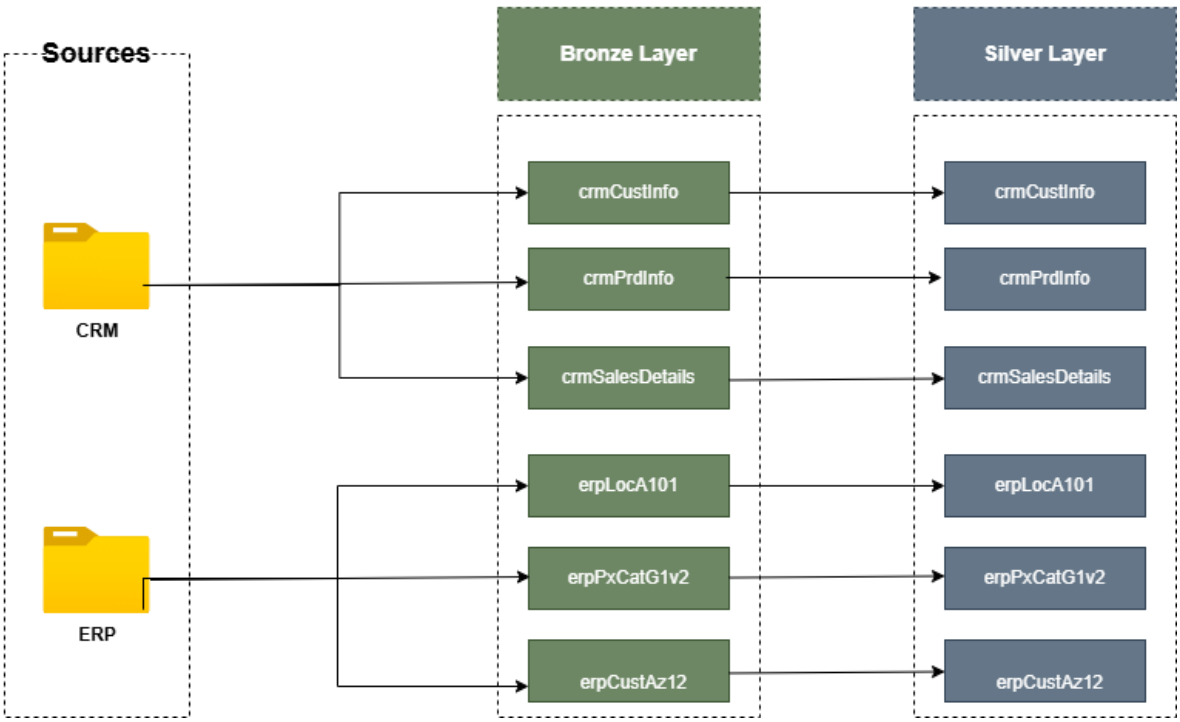
Task 2. Silver Layer

Integration Model

Intergration Model (How the tables are related)



Silver Layer Data Flow Diagram



Data Quality Checks

- Scripts found under the tests folder “checksBronze.sql”

```
==================================================================
Checking silver.crmCustInfo
==================================================================
Check for NULLs or Duplicates in Primary Key

(0 rows affected)
Check for Unwanted Spaces

(0 rows affected)
Data Standardization & Consistency

(2 rows affected)
==================================================================
Checking silver.crmPrdInfo
==================================================================
Check for NULLs or Duplicates in Primary Key

(0 rows affected)
Check for Unwanted Spaces

(0 rows affected)
Check for NULLs or Negative Values in Cost
```

```
(0 rows affected)
Data Standardization & Consistency

(5 rows affected)
Check for Invalid Date Orders (Start Date > End Date)

(0 rows affected)
====================================================================
Checking silver.crmSalesDetails
====================================================================
Check for Invalid Dates

(0 rows affected)
Check for Invalid Date Orders (Order Date > Shipping/Due Dates)

(0 rows affected)
Check Data Consistency: Sales = Quantity * Price

(0 rows affected)
====================================================================
Checking silver.erpCustAz12
====================================================================
Identify Out-of-Range Dates
Expectation: Birthdates between 1900-01-01 and Today. We cannot expect to have a
customer of over 103 years old

(0 rows affected)
Data Standardization & Consistency

(3 rows affected)
====================================================================
Checking silver.erpLocA101
====================================================================
Data Standardization & Consistency

(7 rows affected)
====================================================================
Checking silver.erpPxCatG1v2
====================================================================
Check for Unwanted Spaces

(0 rows affected)
Data Standardization & Consistency

(2 rows affected)

Completion time: 2026-05-08T18:09:01.6736030+02:00
```

	cstID	(No column name)

	cstKey

	cstMaritalStatus
1	Single
2	Married

	prdId	(No column name)

	prdNm

	prdCost

	prdLine
1	Mountain
2	n/a
3	Other S...
4	Road
5	Touring

prdId	dwhCatId	prdKey	prdNm	prdCost	prdLine	prdStartDt	prdEndDt	dwhCreateDate

	slsDueDt

slsOrdNum	slsPrdKey	slsCustId	slsOrderDt	slsShipDt	slsDueDt	slsSales	slsQuantity	slsPrice	dwhCreateDate

slsSales	slsQuantity	slsPrice

	gen
1	n/a
2	Male
3	Fe...

	cntry
1	Australia
2	Canada
3	France
4	Germany
5	n/a
6	United...
7	United...

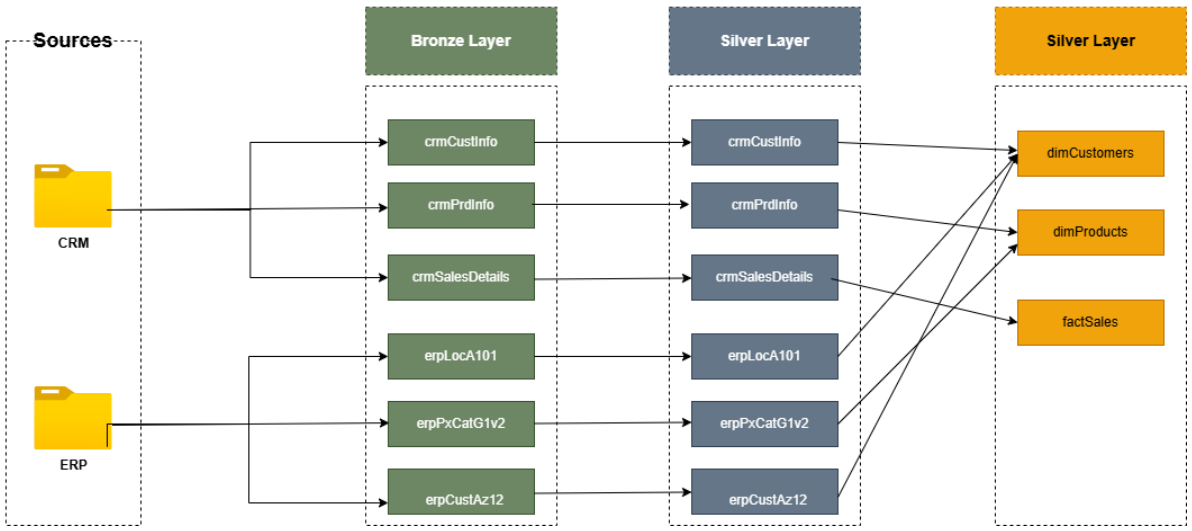
id	cat	subcat	maintenance	dwhCreateDate

	maintenance
1	No
2	Yes

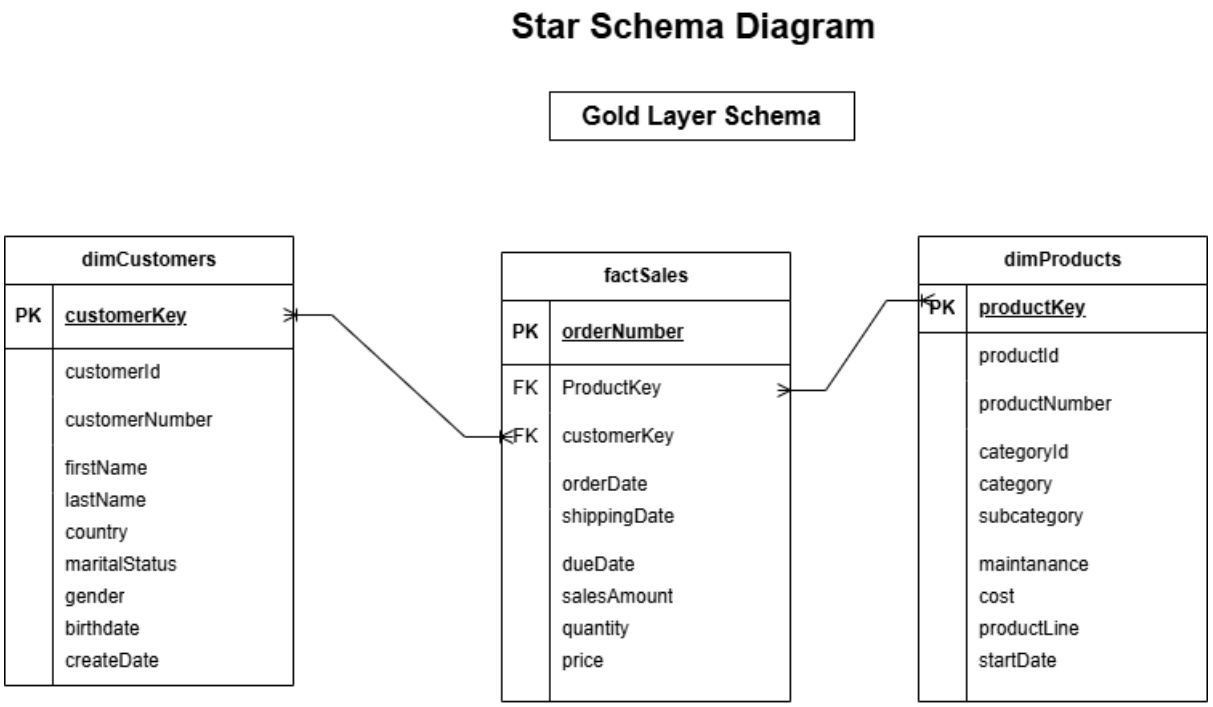
Task 3 Gold Layer

Data Flow Diagram

Data Flow Diagram



Star Schema



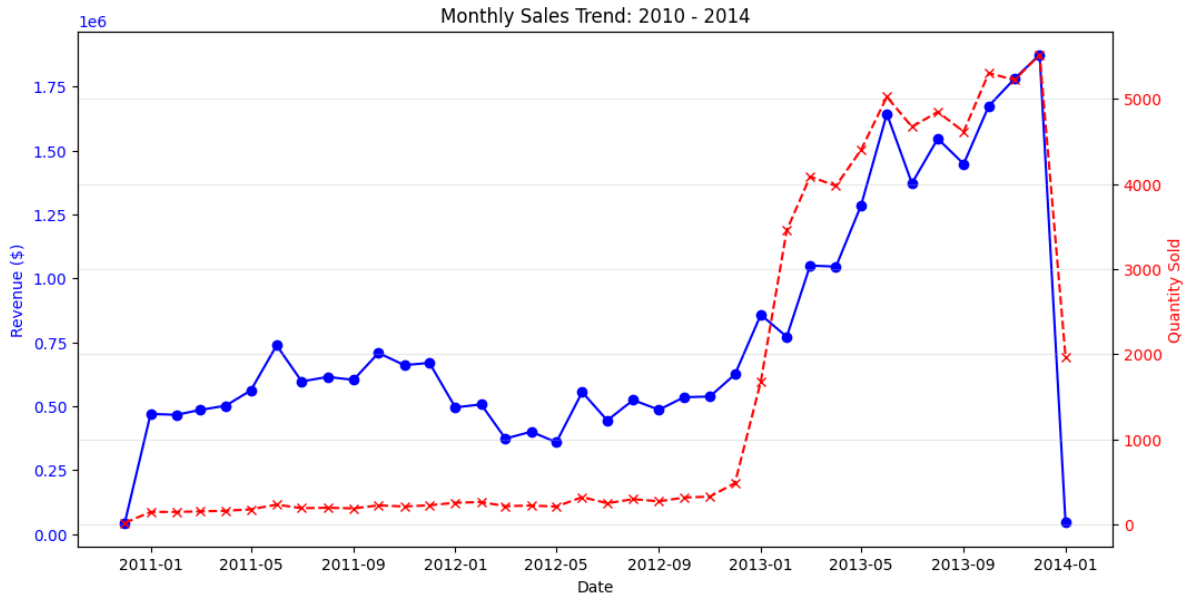
Task 4. Analytical Queries Business Intelligence Reporting

4.1 Total Sales Revenue by Product Category

Category_Name	Total_Sales
Bikes	28316272
Accessories	700262
Clothing	339716

The data shows that Bikes is the primary revenue driver, generating over 28 million in sales. This figure vastly outperforms Accessories and Clothing, which contribute approximately 700,000 and 340,000 respectively. Overall, the business appears heavily reliant on bike sales, as they account for the vast majority of the total volume shown.

4.2 Monthly Sales Trend



The chart shows a significant growth surge in both revenue and quantity sold starting in early 2013, following two years of relatively flat performance. While both metrics track closely during this peak, there is a drastic drop-off for both lines at the beginning of 2014. This suggests a major event, such as a season ending or a data reporting cutoff, rather than a gradual decline.

Year	Month	Revenue	Quantity

2010	12	43419	14
2011	1	469795	144
2011	2	466307	144
2011	3	485165	150
2011	4	502042	157
2011	5	561647	174
2011	6	737793	230
2011	7	596710	188
2011	8	614516	193
2011	9	603047	185
2011	10	708164	221
2011	11	660507	208
2011	12	669395	222
2012	1	495363	252
2012	2	506992	260
2012	3	373478	212
2012	4	400324	219
2012	5	358866	207
2012	6	555142	318
2012	7	444533	246
2012	8	523887	294
2012	9	486149	269
2012	10	535125	313
2012	11	537918	324
2012	12	624454	483
2013	1	857758	1677
2013	2	771218	3454
2013	3	1049732	4087
2013	4	1045860	3979
2013	5	1284456	4400
2013	6	1642948	5025
2013	7	1371595	4673
2013	8	1545910	4848
2013	9	1447324	4616
2013	10	1673261	5304
2013	11	1780688	5224
2013	12	1874128	5520
2014	1	45642	1970

4.3 Top 10 Products by Revenue

productName	productLine	Total_Revenue	Total_Quantity_Sold
Mountain-200 Black- 46	Mountain	1373454	620
Mountain-200 Black- 42	Mountain	1363128	614
Mountain-200 Silver- 38	Mountain	1339394	596
Mountain-200 Silver- 46	Mountain	1301029	580
Mountain-200 Black- 38	Mountain	1294854	582
Mountain-200 Silver- 42	Mountain	1257368	560
Road-150 Red- 48	Road	1205786	337
Road-150 Red- 62	Road	1202208	336
Road-150 Red- 52	Road	1080556	302
Road-150 Red- 56	Road	1055510	295

The data indicates that the Mountain-200 line is the top performer, holding all six of the highest-grossing spots. These mountain bikes significantly outsell the Road-150 models in both total revenue and quantity, with roughly double the unit volume. Specifically, the Black-46 variant of the Mountain-200 is the most successful individual product in this set.

4.4 Customer Segmentation by Country

country	Total_Customers	Total_Orders	Total_Revenue
United States	7482	9230	9162327
Australia	3591	6718	9060172
United King...	1913	3031	3391376
Germany	1780	2484	2894066
France	1810	2484	2643751
Canada	1571	3375	1977738
n/a	337	337	226820

The United States is the leading market, boasting the highest number of customers, orders, and total revenue. Australia is remarkably efficient, generating nearly as much revenue as the U.S. despite having less than half the customer base. Other regions like the UK, Germany, and France form a secondary tier with more moderate, consistent sales figures.

4.5 Average Order Value by Customer Gender

gender	Avg_Order_Value
Female	493
Male	478

The data shows that Female customers have a slightly higher average order value of 493 compared to 478 for Males. This represents a marginal difference of about 3%, indicating relatively consistent spending behavior across both groups. Overall, the target market appears well-balanced in terms of how much each gender spends per transaction.

4.6 Product Subcategory Profitability Analysis

Sub_Category	Total_revenue	Total_Cost	profit_margin_pct
Socks	5112	1704	66.67
Fenders	46662	16968	63.64
Tires and Tubes	244634	90710	62.92
Bike Stands	39591	14691	62.89
Helmets	225435	83733	62.86
Shorts	71330	26494	62.86
Vests	36160	13560	62.50
Gloves	34320	12870	62.50
Bike Racks	39360	14760	62.50
Cleaners	7272	2727	62.50
Hydration Packs	40315	15393	61.82
Bottles and Cages	56993	21770	61.80
Mountain Bikes	9952254	5597922	43.75
Touring Bikes	3844580	2389710	37.84
Road Bikes	14519438	9219075	36.51
Jerseys	173084	133076	23.11
Caps	19710	15330	22.22

The data reveals a striking contrast: while Road and Mountain Bikes generate the highest revenue, they operate on much lower profit margins (36%–43%). In contrast, smaller categories like Socks and Helmets are highly efficient, boasting margins exceeding 60%. This indicates a strategic trade-off between high-volume revenue from bike sales and high-margin profitability from accessories and apparel.

4.7 Shipping Delay Analysis

Category	Avg_Shipping_Delay_Days
Accessories	7
Bikes	7
Clothing	7

Shipping performance is perfectly uniform across the board, with all three categories averaging exactly 7 days of delay. This consistency suggests that logistics bottlenecks are likely tied to general warehouse processing or carrier schedules rather than the specific nature of the products themselves. Because there is no variation, no single category is currently performing better or worse in terms of delivery speed.

Task 5. Technical Documentation and Critical Reflection

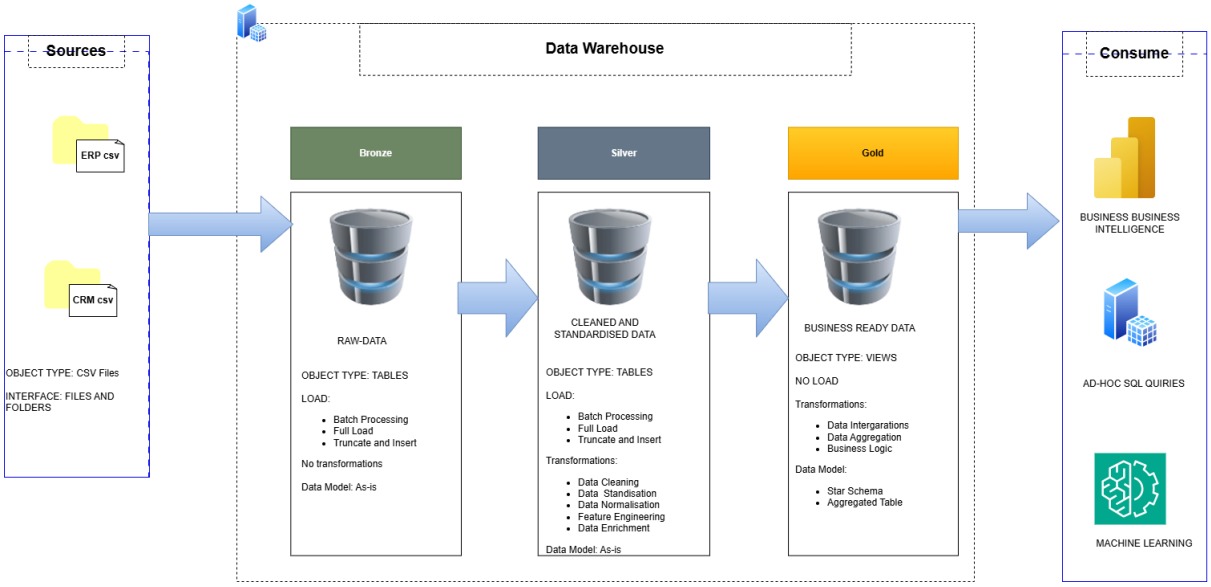
5.1 Architecture Diagram

Data Management Approach

This is the design of how the data will flow and be consumed. In this project, we will design a Data Warehouse for reporting using the Medallion architecture. The

Medallion Architecture is made up of three layers, namely: Bronze, Silver, and Gold, and at each layer, there is a separation of concerns. At the Bronze, we keep the original data as it is from the sources. We transform and clean data at Silver Layer. We build different types of objects for business intelligence and machine learning at the Gold Layer.

Medallion Architecture



5.2 Data Catalogue Gold Layer

View: dimCustomers

This dimension provides a descriptive profile of the customer base.

Column Name	Data Type	Source	Description	Key Type
customerKey	INT	*Generated*	Unique surrogate key for the customer.	SK
customerId	INT	silver.crmCustInfo.cstId	Unique identifier from the CRM system.	Attribute
customerNumber	NVARCHAR(20)	silver.crmCustInfo.cstKey	Business-facing unique customer ID.	Attribute
firstName	NVARCHAR(50)	silver.crmCustInfo.cstFirstname	Customer's first name.	Attribute
lastName	NVARCHAR(50)	silver.crmCustInfo.cstLastname	Customer's last name.	Attribute
country	NVARCHAR(50)	silver.crmCustInfo.cstGndr	Country of residence.	Attribute
maritalStatus	NVARCHAR(10)	silver.crmCustInfo.cstMarital	Marital status (e.g., Single, Married).	Attribute

gender	NVARCHAR(10)	silver.crmCustInfo.cstGndr	Gender of the customer.	Attribute
birthdate	DATE	silver.crmCustInfo.cstBirthdate	Date of birth for age demographics.	Attribute
createDate	DATETIME	silver.crmCustInfo.cstCreateDate	Date the customer record was created.	Attribute

View: dimProducts

This dimension stores details for all items sold, including category hierarchies.

Column Name	Data Type	Source	Description	Key Type
productKey	INT	*Generated*	Unique surrogate key for the product.	SK
productId	INT	silver.cmrPrdInfo.prdId	Unique identifier from the ERP system.	Attribute
productNumber	NVARCHAR(20)	silver.cmrPrdInfo.prdKey	Business-facing product ID / SKU.	Attribute
categoryId	INT	silver.erpPxCatG1v2.catId	ID of the high-level product category.	Attribute
category	NVARCHAR(50)	silver.erpPxCatG1v2.catNm	Name of the primary category (e.g., Bikes).	Attribute
subcategory	NVARCHAR(50)	silver.erpPxCatG1v2.subCatNm	Specific product sub-type (e.g., Mountain).	Attribute
maintanance	NVARCHAR(10)	silver.erpPxCatG1v2.maintenance	Maintenance requirement level.	Attribute
cost	DECIMAL(18,2)	silver.cmrPrdInfo.prdCost	Manufacturing or purchase cost per unit.	Attribute
productLine	NVARCHAR(20)	silver.cmrPrdInfo.prdLine	Marketing line associated with the item.	Attribute
startDate	DATE	silver.cmrPrdInfo.prdStartDt	Date the product was first introduced.	Attribute

View: factSales

The central table containing transactional quantitative data.

Column Name	Data Type	Source	Description	Key Type
orderNumber	NVARCHAR(20)	silver.crmSalesDetails.slsOrdNum	Unique identifier for the sales order.	PK
productKey	INT	dimProducts.productKey	Link to the product dimension.	FK
customerKey	INT	dimCustomers.customerKey	Link to the customer dimension.	FK
orderDate	DATE	silver.crmSalesDetails.slsOrdDt	Date the order was placed.	Attribute
shippingDate	DATE	silver.crmSalesDetails.slsShipDt	Date the order was shipped.	Attribute
dueDate	DATE	silver.crmSalesDetails.slsDueDt	Date the payment is due.	Attribute
salesAmount	DECIMAL(18,2)	silver.crmSalesDetails.slsAmt	Total gross revenue from the line item.	Measure
quantity	INT	silver.crmSalesDetails.slsQty	Number of units sold.	Measure
price	DECIMAL(18,2)	silver.crmSalesDetails.slsPrice	Unit price at the time of sale.	Measure

5.3 ETL Design Decisions and Justifications

During the ETL design process for this Star Schema, the primary goal was to move data from a messy Silver Layer (replicated source systems) to a clean, reporting-ready Gold Layer.

Here are four specific technical decisions made to ensure data integrity and performance:

1. Deduplication using ROW_NUMBER() over DISTINCT

- The Problem: Source CRM data often contains duplicate entries for the same customer (e.g., re-entry of the same cst_id with updated timestamps). Using DISTINCT only removes exact row matches, but doesn't handle logic where you only want the *latest* record.
- The Approach: I utilized ROW_NUMBER() OVER (PARTITION BY cst_id ORDER BY cst_create_date DESC) in the Silver-to-Gold transformation.
- Why: This allows us to select exactly one record per customer (the most recent) and filter out older or erroneous duplicates by selecting only where the row number is 1.
- Code Reference: SELECT * FROM (SELECT *, ROW_NUMBER() OVER (PARTITION BY customerId ORDER BY createDate DESC) as rn FROM silver.crmCustInfo) t WHERE rn = 1;

2. Handling Nulls with ISNULL() for Fixed Defaults

- The Problem: The country and maritalStatus fields in the Customer table frequently contained NULL values, which would break grouping in BI reports (appearing as "Unknown" or blank).

- The Approach: I selected ISNULL(country, 'n/a') for these categorical fields.
- Why: While COALESCE() is more flexible for multiple columns, ISNULL() is slightly more performant in SQL Server for single-value checks and explicitly forces the data type of the replacement value to match the column.
- Code Reference: SELECT ISNULL(country, 'n/a') AS country, ISNULL(maritalStatus, 'n/a') AS maritalStatus FROM silver.crmCustInfo;

---

3. Implementing Surrogate Keys (SK) for Dimensions

- The Problem: The productId and customerId from the source systems are "Natural Keys." If these systems are ever merged or IDs are reused, the relationships in the Data Warehouse would break.
- The Approach: I created productKey and customerKey as Surrogate Keys (identity integers) in the Gold layer.
- Why: This decouples the Data Warehouse from source system changes, improves join performance (using integers instead of strings), and allows for historical tracking (SCD Type 2) in the future.
- Code Reference: CREATE TABLE gold.dim_products (productKey INT IDENTITY(1,1) PRIMARY KEY, productId INT, ...);

4. Schema Binding for Date Calculations

- The Problem: Calculating order_to_ship lag or ages during runtime slows down large fact table queries.
- The Approach: I pre-calculated essential dates and flags during the ETL load into fact_sales.
- Why: By doing the "heavy lifting" (like calculating DATEDIFF between orderDate and shippingDate) during the ETL process rather than in a Power BI measure, we reduce the computational load on the end-user report.
- Code Reference: INSERT INTO gold.fact_sales SELECT ..., DATEDIFF(day, orderDate, shippingDate) AS ShippingDelayDays FROM silver.erp_sales_hdr;

5.4 Limitations and Proposed Enhancements

The current architecture serves as a solid foundation, but it is currently a "static" snapshot that lacks the robustness required for a production-grade enterprise environment.

Absence of a Dedicated Date Dimension (dimDate)

- The Limitation: Currently, time-based analysis relies on raw date columns in factSales. This makes it impossible to filter by complex business logic like fiscal quarters, public holidays, or "working days vs. weekends" without writing complex SQL every time.

- Proposed Enhancement: Generate a static gold.dimDate table covering a 20-year range (e.g., 2010–2030).
- Technical Implementation: Populate the table using a SQL recursive CTE to generate one row per day. Include pre-calculated attributes like IsWeekend, MonthName, FiscalYear, and DayOfWeek. In the fact_sales view, replace raw dates with DateKey integers (YYYYMMDD) to optimize join performance.

Lack of Incremental Load Logic

- The Limitation: The current ETL process likely performs a "Full Truncate and Load," meaning every record is reprocessed daily. As the data grows from thousands to millions of rows, the processing window will eventually exceed the allotted time, causing system downtime.
- Proposed Enhancement: Implement a "Watermark" or "Delta" loading pattern using a LastUpdated timestamp.
- Technical Implementation:
    - Add a sysLoadDate column to all Gold tables.
    - Create a metadata table to store the LastSuccessfulRunDateTime.
    - Modify the ETL scripts to only fetch records from Silver where silverModifiedDate> LastSuccessfulRunDateTime.
    - Use the MERGE statement in SQL Server to UPDATE existing records and INSERT new ones in a single pass.

No Slowly Changing Dimension (SCD) Handling

- The Limitation: If a customer moves from the US to Australia, the current system simply overwrites the old value. This "corrupts" historical reports, as past US orders will now incorrectly appear as Australian revenue in retrospective analyses.
- Proposed Enhancement: Implement SCD Type 2 for the dimCustomers and dimCroducts tables.
- Technical Implementation: Add three audit columns to the dimensions: validFrom, valid_to, and isCurrent (boolean). When a change is detected in the source, the ETL

will expire the old record by setting isCurrent = 0 and insert a new record with the updated details, preserving the historical trail.